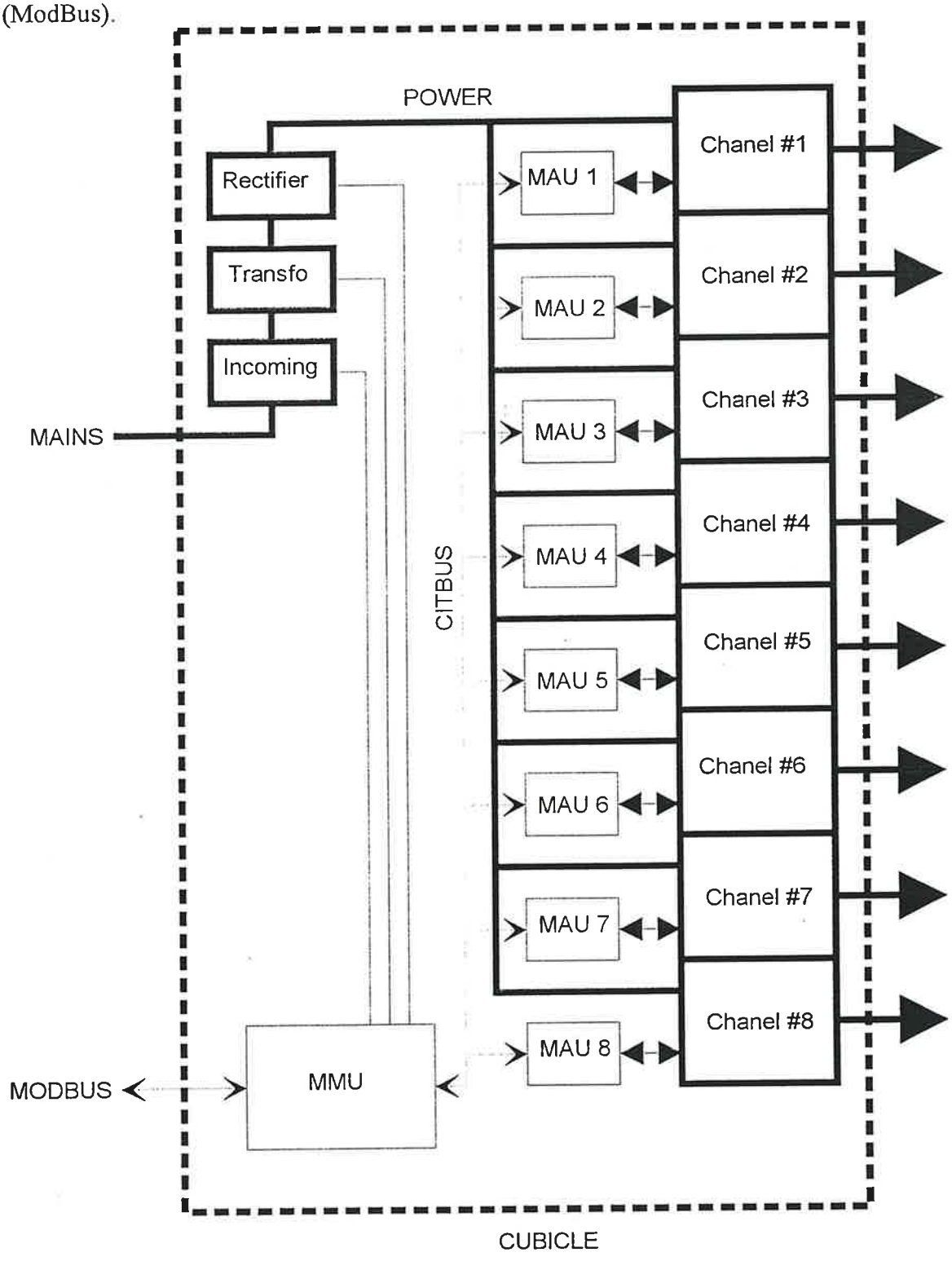
2. REMOTE MONITORING AND CONTROL

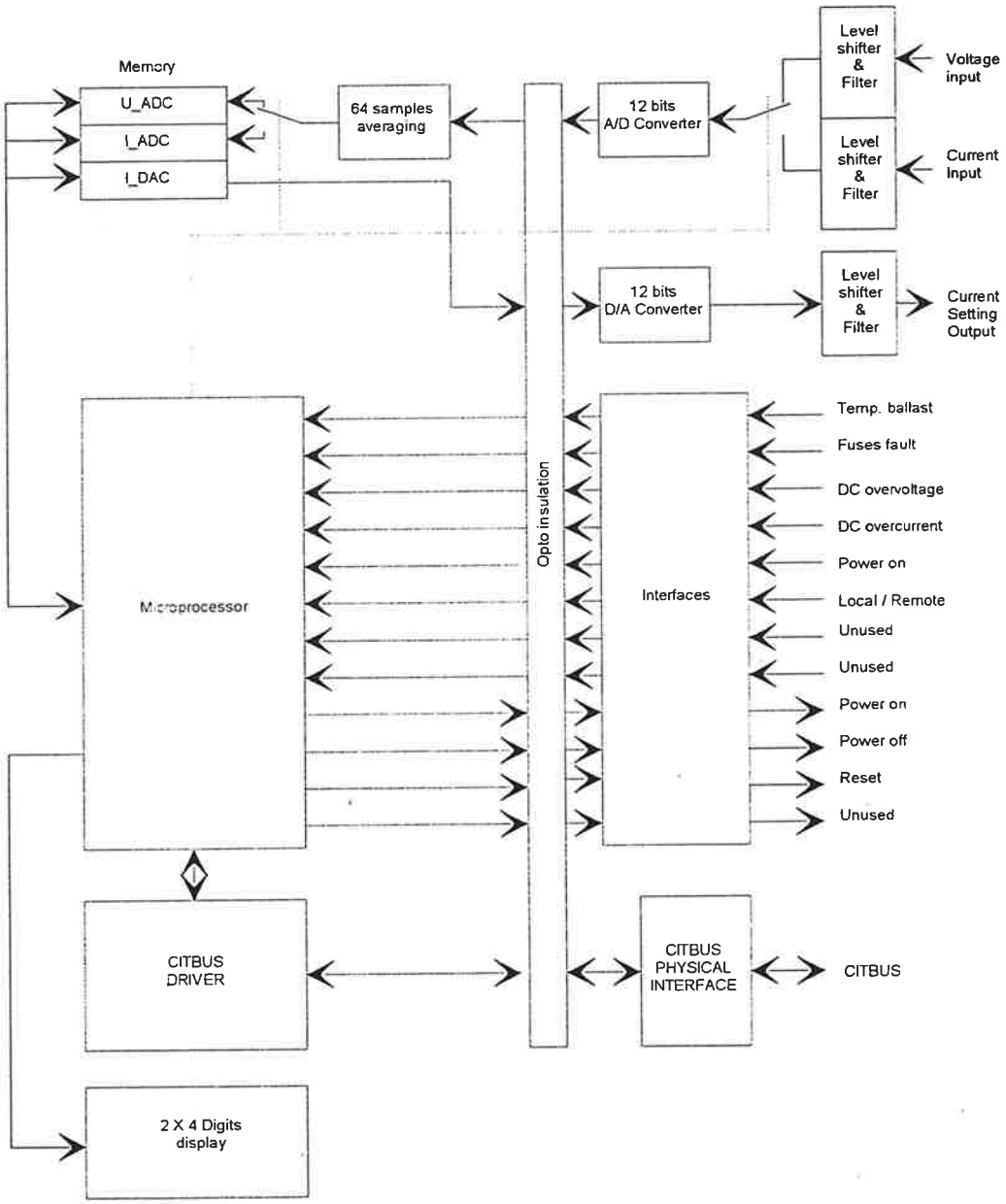
2.1. Global Architecture

The monitoring and control functions are achieved with a distributed system. Each chanel in a power supply is controlled by a Monitoring Auxiliary Unit (MAU). The 8 MAUs are connected to a Main Monitoring Unit (MMU) via a serial communication link (CITBus). On the other side, MMU is connected to the general control system via an other serial link (ModBus).



2.2. Monitoring Auxiliary Unit MAU

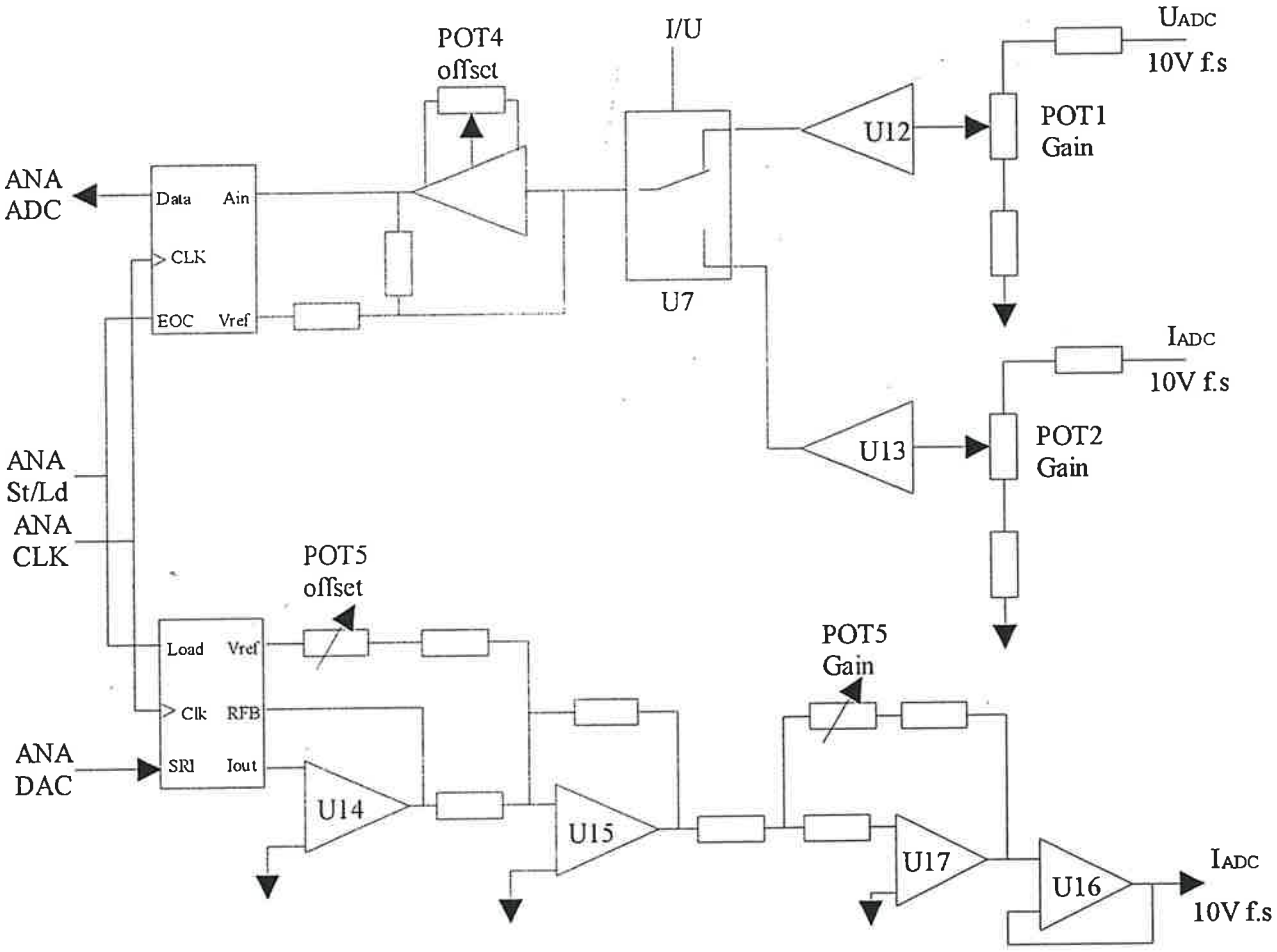
MAU is the level #0 of the monitoring system. It acts as an interface between the AL252 power supply board and the communication bus. MAU insures the load voltage and current analog to digital conversion, reads the status and drives the remote controls of the AL252 board. It also displays the load voltage and current on a front panel display.



Eah MAU is composed of three cards :

- the CPU card including :
  - the processor core,
  - the 8 opto insulated logical inputs,
  - the 4 opto insulated logical outputs,
  - the CITBUS interface.
- the DISPLAY card including 8 digits and their drivers.
- the ANALOG card including :
  - a voltage and a current input filter,
  - a two inputs multiplexor for the voltage and the current input,
  - a 12 bits analog to digital converter (MAX170),
  - a 12 bits digital to analog converter(MAX 543),
  - a buffer for the current setting output,
  - a set of optos for insulation of the card.

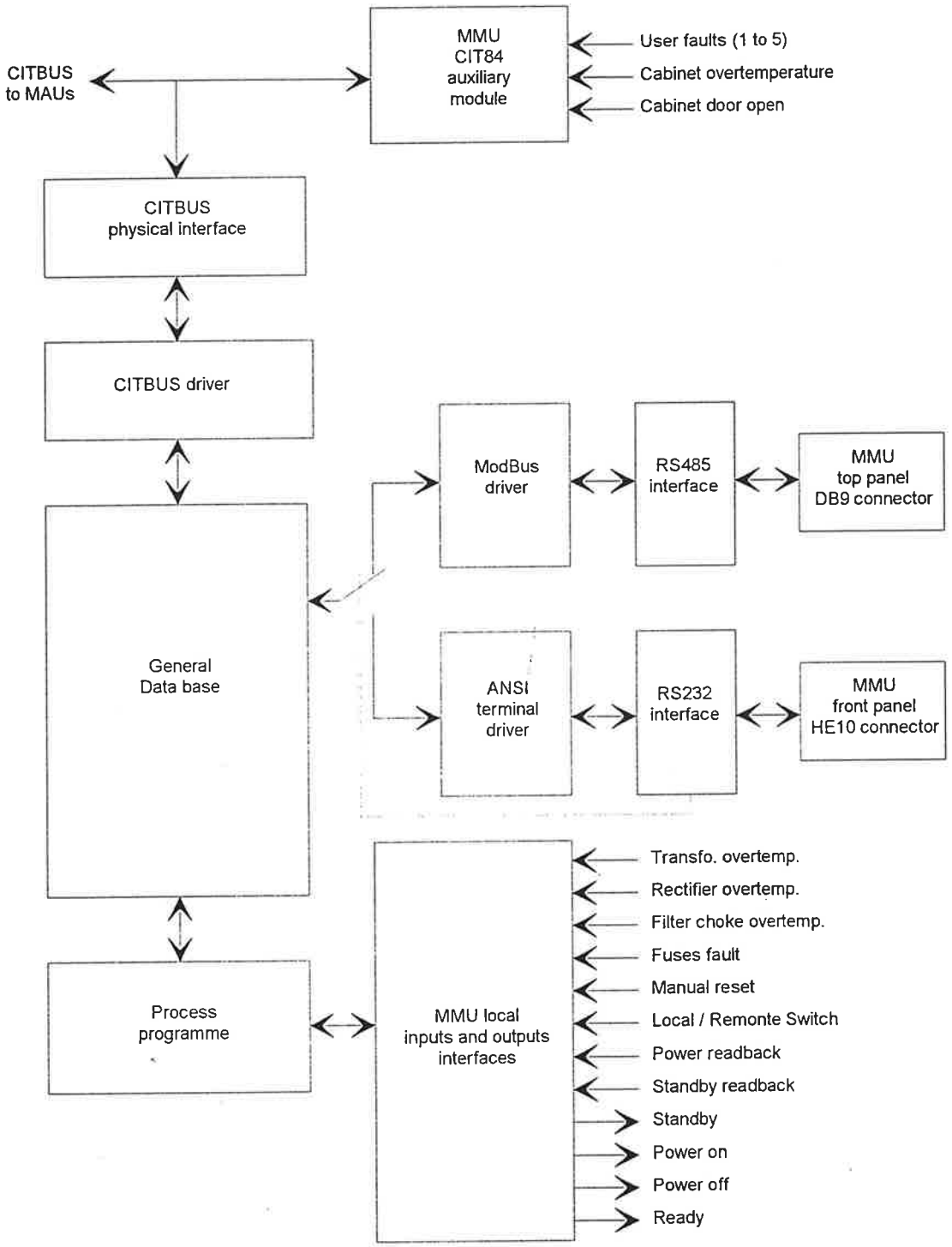
For each chanel, an offset and a full scale factor setting is available (the offset setting is common fr the two inputs). The following diagram is showing the different setting potentiometers.



2.3. Internal communication bus CITBus

All the 8 MAUs are connected to the MMU module via a serial bus using an HAZEMEYER standart physical interface and protocole named CITBUS.
The physical interface is a bidirectionnal current loop.
The protocole is a master / slave type. MMU is the master and MAUs are 8 slaves (the CIT84 MMU auxiliary module is also a slave). The master sends successively a READALL request to each slave which answers by sending the status of its logical and analogical inputs and outputs. It's the polling phase. This polling phase can be interrupted by a WRITEOUT request when the master have to change the value of an output in a slave.

2.4. Main Monitoring Unit MMU



MMU is the central element of the monitoring system. It insures the following functions :

- the data communications between the power supply cubicle and the higher level control system via ModBus,
- the ModBus data base updating,
- the cubicle operating,
- the CITBUS management,
- the local terminal management.

ModBus data base structure and operating state diagrams are visible on appendix.

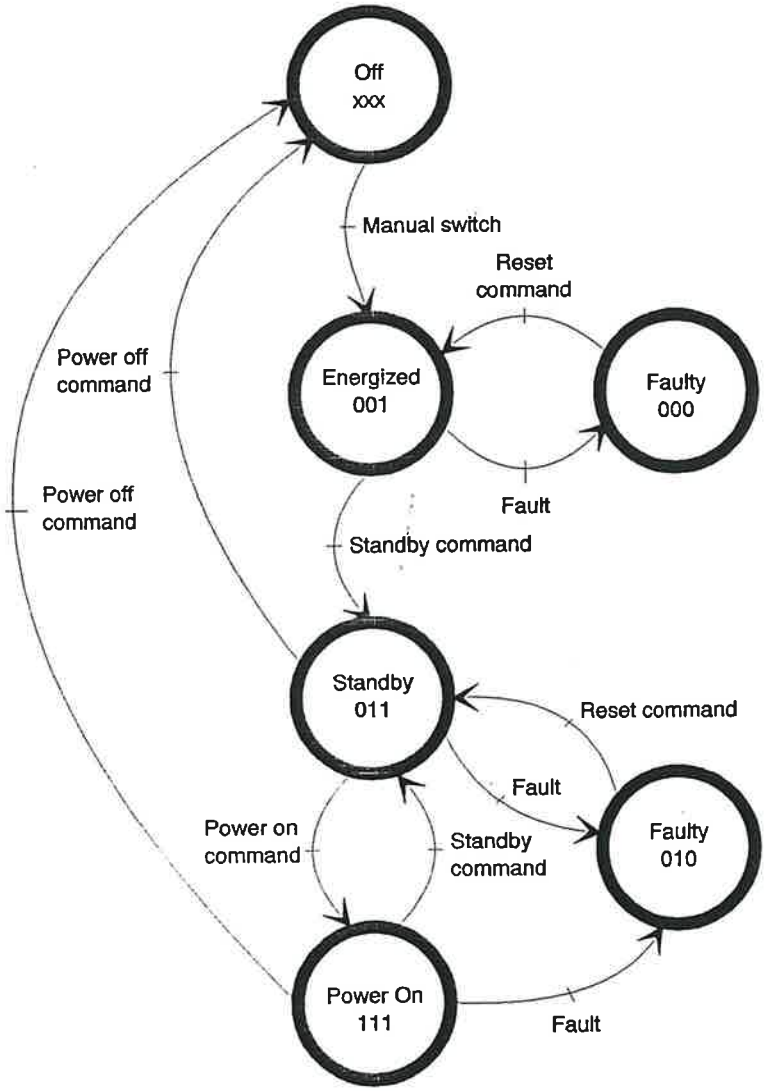
3. AL255: SUPPLY CARD 4x ±15V: (See drawing n° 131110.A.318)

These card supply in electronic crate four AL252

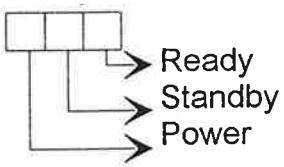
AL250 OPERATING DIAGRAMS AND MMU DATABASE ORGANISATION.

HAZEMEYER

General state diagram



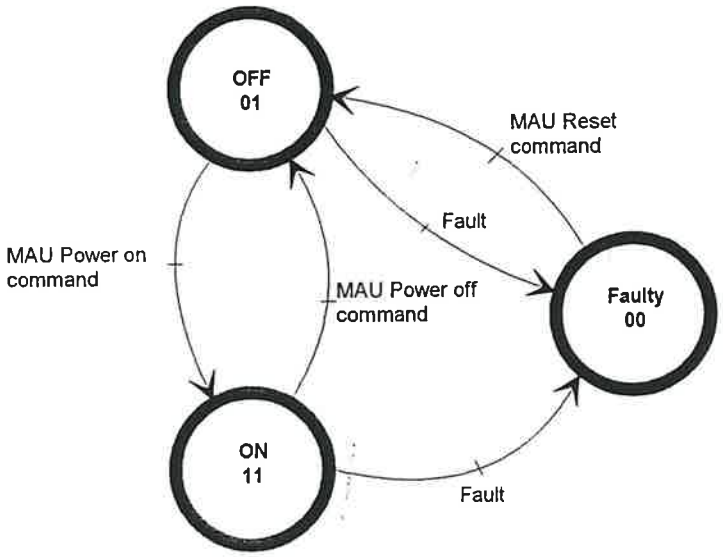
State descriptor :

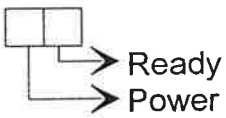


Standby, Power On, Power Off and Reset are Local or Remote commands

HAZEMEYER

Chanel state diagram



State descriptor : 

Power ON, Power OFF and Reset are Local or Remote commands

WRITE BIT AREA

Word Address	Bit Address	
$0000	$0000	Stand By remote command
$0000	$0001	Power On remote command
$0000	$0002	Power Off remote command
$0000	$0003	Remote Reset
$0000	$0004	Active energy reset
$0000	$0005	Reactive energy reset
$0000	$0006 to $000F	Reserved
$0001 to $000F	$0010 to $00FF	Reserved
$0010	$0100	MAU0 Power On remote commande
$0010	$0101	MAU0 Power Off remote commande
$0010	$0102	MAU0 remote Reset
$0010	$0003 to $000F	Reserved
$0011	$0110 to $010F	Reserved
$0012	$0120	MAU1 Power On remote commande
$0012	$0121	MAU1 Power Off remote commande
$0012	$0122	MAU1 remote Reset
$0012	$0123 to $012F	Reserved
$0013	$0130 to $013F	Reserved
$0014	$0140	MAU2 Power On remote commande
$0014	$0141	MAU2 Power Off remote commande
$0014	$0142	MAU2 remote Reset
$0014	$0143 to $014F	Reserved
$0015	$0150 to $015F	Reserved
$0016	$0160	MAU3 Power On remote commande
$0016	$0161	MAU3 Power Off remote commande
$0016	$0162	MAU3 remote Reset
$0016	$0163 to $016F	Reserved
$0017	$0170 to $017F	Reserved
$0018	$0180	MAU4 Power On remote commande
$0018	$0181	MAU4 Power Off remote commande
$0018	$0182	MAU4 remote Reset
$0018	$0183 to $018F	Reserved
$0019	$0190 to $019F	Reserved
$001A	$01A0	MAU5 Power On remote commande
$001A	$01A1	MAU5 Power Off remote commande
$001A	$01A2	MAU5 remote Reset
$001A	$01A3 to $01A	Reserved
$001B	$01B0 to $01B	Reserved
$001C	$01C0	MAU6 Power On remote commande
$001C	$01C1	MAU6 Power Off remote commande
$001C	$01C2	MAU6 remote Reset
$001C	$01C3 to $01C	Reserved
$001D	$01D0 to $01D	Reserved
$001E	$01E0	MAU7 Power On remote commande
$001E	$01E1	MAU7 Power Off remote commande
$001E	$01E2	MAU7 remote Reset
$001E	$01E3 to $01E	Reserved
$001F	$01F0 to $01FF	Reserved

READ BIT AREA

Word Address	Bit Address	
$0020	$0200	Transfo over-temperature
$0020	$0201	Rectifier over-temperature
$0020	$0202	Filter Chocke over-temperature
$0200	$0203	Fuses fault
$0020	$0204	Earth fault
$0020	$0205	User fault #1
$0020	$0206	User fault #2
$0020	$0207	User fault #3
$0020	$0208	User fault #4
$0020	$0209	User fault #5
$0020	$020A	Door open
$0020	$020B	Cabinet over-temperature
$0020	$020C	Reserved
$0020	$020D	Reserved
$0020	$020E	Reserved
$0020	$020F	Reserved
$0021	$0210	Ready status
$0021	$0211	Stand by status
$0021	$0212	Power status
$0021	$0213	Local / Remote status
$0021	$0214	Reset push button status
$0021	$0215	Reserved
$0021	$0216	Reserved
$0021	$0217	Reserved
$0021	$0218	Reserved
$0021	$0219	Reserved
$0021	$021A	Reserved
$0021	$021B	Reserved
$0021	$021C	Reserved
$0021	$021D	Reserved
$0021	$021E	Reserved
$0021	$021F	Reserved
$0022 to $0029	$0220 to $029F	Reserved
$0030	$0300	MAU0 Ballast over-temperature
$0030	$0301	MAU0 Fuses fault
$0030	$0302	MAU0 DC over-voltage
$0030	$0303	MAU0 DC over-current
$0030	$0304	MAU0 Communication failure
$0030	$0305 to $030F	Reserved
$0031	$0310	MAU0 Power status
$0031	$0311 to $031F	Reserved
$0032	$0320	MAU1 Ballast over-temperature
$0032	$0321	MAU1 Fuses fault
$0032	$0322	MAU1 DC over-voltage
$0032	$0323	MAU1 DC over-current
$0032	$0324	MAU1 Communication failure
$0032	$0325 to $032F	Reserved
$0033	$0330	MAU1 Power status
$0033	$0331 to $033F	Reserved
$0034	$0340	MAU2 Ballast over-temperature
$0034	$0341	MAU2 Fuses fault
$0034	$0342	MAU2 DC over-voltage

$0034	$0343	MAU2 DC over-current
$0034	$0344	MAU2 Communication failure
$0034	$0345 to $034F	Reserved
$0035	$0350	MAU2 Power status
$0035	$0351 to $035F	Reserved
$0036	$0360	MAU3 Ballast over-temperature
$0036	$0361	MAU3 Fuses fault
$0036	$0362	MAU3 DC over-voltage
$0036	$0363	MAU3 DC over-current
$0036	$0364	MAU3 Communication failure
$0036	$0365 to $036F	Reserved
$0037	$0370	MAU3 Power status
$0037	$0371 to $037F	Reserved
$0038	$0380	MAU4 Ballast over-temperature
$0038	$0381	MAU4 Fuses fault
$0038	$0382	MAU4 DC over-voltage
$0038	$0383	MAU4 DC over-current
$0038	$0384	MAU4 Communication failure
$0038	$0385 to $038F	Reserved
$0039	$0390	MAU4 Power status
$0039	$0391 to $039F	Reserved
$003A	$03A0	MAU5 Ballast over-temperature
$003A	$03A1	MAU5 Fuses fault
$003A	$03A2	MAU5 DC over-voltage
$003A	$03A3	MAU5 DC over-current
$003A	$03A4	MAU5 Communication failure
$003A	$03A5 to $03A	Reserved
$003B	$03B0	MAU5 Power status
$003B	$03B1 to $03B	Reserved
$003C	$03C0	MAU6 Ballast over-temperature
$003C	$03C1	MAU6 Fuses fault
$003C	$03C2	MAU6 DC over-voltage
$003C	$03C3	MAU6 DC over-current
$003C	$03C4	MAU6 Communication failure
$003C	$03C5 to $03C	Reserved
$003D	$03D0	MAU6 Power status
$003D	$03D1 to $03D	Reserved
$003E	$03E0	MAU7 Ballast over-temperature
$003E	$03E1	MAU7 Fuses fault
$003E	$03E2	MAU7 DC over-voltage
$003E	$03E3	MAU7 DC over-current
$003E	$03E4	MAU7 Communication failure
$003E	$03E5 to $03E	Reserved
$003F	$03F0	MAU7 Power status
$003F	$03F1 to $03FF	Reserved

WRITE WORD AREA

Word Address	Bit Address	
$0060	----------	MAU0 Current setting
$0061	----------	MAU1 Current setting
$0062	----------	MAU2 Current setting
$0063	----------	MAU3 Current setting
$0064	----------	MAU4 Current setting
$0065	----------	MAU5 Current setting
$0066	----------	MAU6 Current setting
$0067	----------	MAU7 Current setting
$0068	----------	Reserved
$0069	----------	Reserved
$006A	----------	Reserved
$006B	----------	Reserved
$006C	----------	Reserved
$006C	----------	Reserved
$006D	----------	Reserved
$006E	----------	Reserved
$0070	----------	Reserved
$0071	----------	Reserved
$0072	----------	Reserved
$0073	----------	Reserved
$0074	----------	Reserved
$0075	----------	Reserved
$0076	----------	Reserved
$0077	----------	Reserved
$0078	----------	Reserved
$0079	----------	Reserved
$007A	----------	Reserved
$007B	----------	Reserved
$007C	----------	Reserved
$007D	----------	Reserved
$007E	----------	Reserved
$007F	----------	Reserved

READ WORD AREA

Word Address	Bit Address	
$0080	----------	Mains phase #1 simple voltage
$0081	----------	Mains phase #2 simple voltage
$0082	----------	Mains phase #3 simple voltage
$0083	----------	Mains average simple voltage
$0084	----------	Mains phase #1 current
$0085	----------	Mains phase #2 current
$0086	----------	Mains phase #3 current
$0087	----------	Mains average current
$0088	----------	Mains phase #1 active power
$0089	----------	Mains phase #2 active power
$008A	----------	Mains phase #3 active power
$008B	----------	Mains total active power
$008C	----------	Mains phase #1 reactive power
$008D	----------	Mains phase #2 reactive power
$008E	----------	Mains phase #3 reactive power
$008F	----------	Mains total reactive power
$0090	----------	Mains phase #1 apparent power
$0091	----------	Mains phase #2 apparent power
$0092	----------	Mains phase #3 apparent power
$0093	----------	Mains total apparent power
$0094	----------	Mains phase #1 power factor
$0095	----------	Mains phase #2 power factor
$0096	----------	Mains phase #3 power factor
$0097	----------	Mains average power factor
$0098	----------	Mains active energy (MSW)
$0099	----------	Mains active energy (LSW)
$009A	----------	Mains reactive energy (MSW)
$009B	----------	Mains reactive energy (LSW)
$009C	----------	Earth fault current
$009D	----------	Mains frequency
$009E	----------	Reserved
$009F	----------	Reserved
$00A0	----------	MAU0 output current
$00A1	----------	MAU0 output voltage
$00A2	----------	Reserved
$00A3	----------	Reserved
$00A4	----------	MAU1 output current
$00A5	----------	MAU1 output voltage
$00A6	----------	Reserved
$00A7	----------	Reserved
$00A8	----------	MAU2 output current
$00A9	----------	MAU2 output voltage
$00AA	----------	Reserved
$00AB	----------	Reserved
$00AC	----------	MAU3 output voltage
$00AD	----------	MAU3 output current
$00AE	----------	Reserved
$00AF	----------	Reserved
$00B0	----------	MAU4 output voltage
$00B1	----------	MAU4 output current
$00B2	----------	Reserved
$00B3	----------	Reserved

$00B4	----------	MAU5 output voltage
$00B5	----------	MAU5 output current
$00B6	----------	Reserved
$00B7	----------	Reserved
$00B8	----------	MAU6 output voltage
$00B9	----------	MAU6 output current
$00BA	----------	Reserved
$00BB	----------	Reserved
$00BC	----------	MAU7 output voltage
$00BD	----------	MAU7 output current
$00BE	----------	Reserved
$00BF	----------	Reserved
$00C0	----------	General Faults (=$0020)
$00C1	----------	General statuses (=$0021)
$00C2	----------	MAU Ready
$00C3	----------	MAU Power Statuses
$00C4	----------	MAU0 output current (=$00A0)
$00C5	----------	MAU1 output current (=$00A4)
$00C6	----------	MAU2 output current (=$00A8)
$00C7	----------	MAU3 output current (=$00AC)
$00C8	----------	MAU4 output current (=$00B0)
$00C9	----------	MAU5 output current (=$00B4)
$00CA	----------	MAU6 output current (=$00B8)
$00CB	----------	MAU7 output current (=$00BC)

MAU Ready description :
bits 0 to 7 indicate if the corresponding MAU is Ready (1) or Faulty (0)

MAU Power Statuses description :
bits 0 to 7 indicate if the corresponding MAU is On (1) or Off (0)

WRITE BIT AREA

Adresse	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0	
0000											rE-Reset	aE-Reset	Reset	PowerOFF	PowerON	StandBy	General
1																	
2																	
3																	
4																	
5																	
6																	
7																	
8																	
9																	
A																	
B																	
C																	
D																	
E																	
F																	
0010														Reset	PowerOFF	PowerON	MAV 0
1																	
2														Reset	PowerOFF	PowerON	MAV 1
3																	
4														Reset	PowerOFF	PowerON	MAV 2
5																	
6														Reset	PowerOFF	PowerON	MAV 3
7																	
8														Reset	PowerOFF	PowerON	MAV 4
9																	
A														Reset	PowerOFF	PowerON	MAV 5
B																	
C														Reset	PowerOFF	PowerON	MAV 6
D																	
E														Reset	PowerOFF	PowerON	MAV 7
F																	

READ BIT AREA

Adresse	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0	
0020					Ø Cubinet	Door Open	User 5	User 4	User 3	User 2	User 1	Earth Fault	Fuse Fault	Ø Chocke	Ø Rect.	Ø Transfo	Gener.
1												Reset	Loc./Rem	Power	Standby	Ready	
2																	
3																	
4																	
5																	
6																	
7																	
8																	
9																	
A																	
B																	
C																	
D																	
E																	
F																	
2030												Com. Fail P.	Dc-OverC	DC-OverV	Fuse Fault	Ø Ballast	MAV Ø
1															Loc./Rem.	Power	
2												Com. Fail.	DC OverC	DC overV	Fuse Fault	Ø Ballast	MAV 1
3															Loc./Rem.	Power	
4												Com. Fail P.	DC OverC	DC OverV	Fuse Fault	Ø Ballast	MAV 2
5															Loc./Rem.	Power	
6												Com. Fail.	DC OverC	DC OverV	Fuse Fault	Ø Ballast	MAV 3
7															Loc./Rem.	Power	
8												Com. Fail.	DC_OverC	DC OverV	Fuse Fault	Ø Ballast	MAV 4
9															Loc./Rem.	Power	
A												Com. Fail.	DC OverC	DC OverV	Fuse Fault	Ø Ballast	MAV 5
B															Loc./Rem.	Power	
C												Com. Fail.	DC OverC	DC OverV	Fuse Fault	Ø Ballast	MAV 6
D															Loc./Rem.	Power	
E												Com. Fail.	DC OverC	DC OverV	Fuse Fault	Ø Ballast	MAV 7
F															Loc./Rem.	Power.	

WRITE WORD AREA (1/2)

Adresse	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
0040																
1																
2																
3																
4																
5																
6																
7																
8																
9																
A																
B																
C																
D																
E																
F																
0050																
1																
2																
3																
4																
5																
6																
7																
8																
9																
A																
B																
C																
D																
E																
F																

Reserved

WRITE WORD AREA (2/2)

Adresse	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
0060	—	—	—	—	—	MAU Ø Current Setting				—	—	—	—	—	—	—
1	—	—	—	—	—	MAU 1 Current Setting				—	—	—	—	—	—	—
2	—	—	—	—	—	MAU 2 Current Setting				—	—	—	—	—	—	—
3	—	—	—	—	—	MAU 3 Current Setting				—	—	—	—	—	—	—
4	—	—	—	—	—	MAU 4 Current Setting				—	—	—	—	—	—	—
5	—	—	—	—	—	MAU 5 Current Setting				—	—	—	—	—	—	—
6	—	—	—	—	—	MAU 6 Current Setting				—	—	—	—	—	—	—
7	—	—	—	—	—	MAU 7 Current Setting				—	—	—	—	—	—	—
8																
9																
A																
B																
C																
D																
E																
F																
0070																
1																
2																
3																
4																
5																
6																
7																
8	—	—	—	—		Reserved										
9																
A																
B																
C																
D																
E																
F																

READ WORD AREA (1/3)

Adresse	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
0080								U1								
1								U2								
2								U3								
3							U average									
4								I1								
5								I2								
6								I3								
7							I average									
8								P1								
9								P2								
A								P3								
B							P total									
C								Q1								
D								Q2								
E								Q3								
F							Q Total									
0090								S1								
1								S2								
2								S3								
3							S Total									
4								PF1								
5								PF2								
6								PF3								
7							PF Average									
8							Active Energy (MSW)									
9							Active Energy (LSW)									
A							Reactive Energy (MSW)									
B							Reactive Energy (MSW)									
C							Earth Fault Current									
D							Frequency									
E																
F																

READ WORD AREA (2/3)

Adresse	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
00A0							MAU0	Current Readout								
1							MAU0	Voltage Readout								
2																
3																
4							MAU1	Current Readout								
5							MAU1	Voltage Readout								
6																
7																
8							MAU2	Current Readout								
9							MAU2	Voltage Readout								
A																
B																
C							MAU3	Current Readout								
D							MAU3	Voltage Readout								
E																
F																
00B0							MAU4	Current Readout								
1							MAU4	Voltage Readout								
2																
3																
4							MAU5	Current Readout								
5							MAU5	Voltage Readout								
6																
7																
8							MAU6	Current Readout								
9							MAU6	Voltage Readout								
A																
B																
C							MAU7	Current Readout								
D							MAU7	Voltage Readout								
E																
F																

READ WORD AREA (3/3)

Adresse	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
00C0						General		Fun Pts	(see	$0020)						
C1						General		Statuses	(see	$0021)						
C2									Ready7	Ready6	Ready5	Ready4	Ready3	Ready2	Ready1	Ready0
C3									Power7	Power6	Power5	Power4	Power3	Power2	Power1	Power0
C4						MAU0 Current Readout										
C5						MAU1 Current Readout										
C6						MAU2 Current Readout										
C7						MAU3 Current Readout										
C8						MAU4 Current Readout										
C9						MAU5 Current Readout										
CA						MAU6 Current Readout										
CB						MAU7 Current Readout										
CC																
CD																
CE																
CF																
00D0																
D1																
D2																
D3																
D4																
D5																
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DD																
DE																
DF																